Cybersecurity Basics 🔐

Junior Cybersecurity Badge — Girls Build Bots

Presented by your robotics mentors!

How does information actually travel? 🌐

Today's layer is the **how-it-works** layer:

- Networks 🕸️
- Messages hopping between computers 💬
- Why data is **valuable** 💎

The Human Internet 

We're building a network — out of you.

1. **Senders** write a short message on a card
2. Seal it in an envelope, addressed across the room
3. It travels **hand to hand through "routers"** to get there

Routers = the Scouts in the middle.

Round 2: the snoop 👀

One router is a **snoop**. Unsealed cards get peeked at.

Debrief: on the internet, messages hop through many computers. Anyone in the path could look.

Which is why some messages need protecting.

Secret Codes 🔑

The solution to the snoop: write your message in a **cipher** before you send it.

1. Encode with the substitution sheet
2. Send it
3. Your partner decodes it

The snoop gets a coded card — and can't read it.

That’s encryption 

A cipher is the core idea of **encryption**.

Encryption is the internet’s envelope.

What’s your data worth? 

Sort the cards. Which would a stranger want most — and why?

password > address > school > favorite song

Your digital footprint 👣

The trail of information you leave online **adds up**.
Here's the honest version:

Data is valuable, so people try to collect it.
That's exactly why we **choose** what to share.

Interland round 🎮

Play a realm that reinforces **share with care**.

Robot moment 

Our robot and its **driver station** talk over a network during matches.

The field's network is locked down so **no one else** can send our robot commands.

Closing circle 🎯

Name: 1. One thing you learned about **how information travels** 2. One piece of data **you'll guard**

What you accomplished ✅

- Explained how a message travels through a network of computers
- Explained why some information needs protecting on the way
- Encoded and decoded a message with a cipher
- Ranked personal data by value — and defended your reasoning

Keep going at home! 🏠

- **beinternetawesome.withgoogle.com** — Interland (free)
- **cyber.org** — free cybersecurity lessons
- Ask your troop leader about the next Cybersecurity badge!

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This is how the program grows — and how the next troop and team get better materials.